

Question 1

1.1. $f(x) = 3x^2 - 2x$

$$\begin{aligned} f(x+h) &= 3(x+h)^2 - 2(x+h) \\ &= 3(x^2 + 2xh + h^2) - 2x - 2h \\ &= 3x^2 + 6xh + 3h^2 - 2x - 2h \end{aligned}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{\overset{\checkmark f(x+h)}{3x^2 + 6xh + 3h^2 - 2x - 2h} - (3x^2 - 2x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h} \quad \checkmark \text{HCF}$$

$$= \lim_{h \rightarrow 0} 6x + 3h - 2$$

$$= 6x + 3(0) - 2 \quad \checkmark$$

$$= 6x - 2 \quad \checkmark \quad (5)$$

$$1.2.1. \frac{d}{dx} [(3x-4)^2]$$

$$\frac{d}{dx} [9x^2 - 24x + 16] \checkmark$$

$$= 18x - 24 \checkmark$$

(3)

$$2.2. \quad g(x) = 8\sqrt{x} - \frac{2}{x^3}$$

$$g(x) = 8x^{\frac{1}{2}} - 2x^{-3} \checkmark$$

$$g'(x) = 4x^{-\frac{1}{2}} + 6x^{-4} \checkmark$$

(3)

$$2.3. \quad s = -3t^4 + pt - q$$

$$\frac{ds}{dt} = -12t^3 + p \checkmark$$

(2)

$$3. \quad h(x) = x^3 - 5x^2 + 3x - 1$$

$$h(1) = (1)^3 - 5(1)^2 + 3(1) - 1$$

$$= -2 \checkmark$$

$$\therefore (1; -2)$$

$$h'(x) = 3x^2 - 10x + 3 \checkmark$$

$$h'(1) = 3(1)^2 - 10(1) + 3$$

$$= -4 \checkmark$$

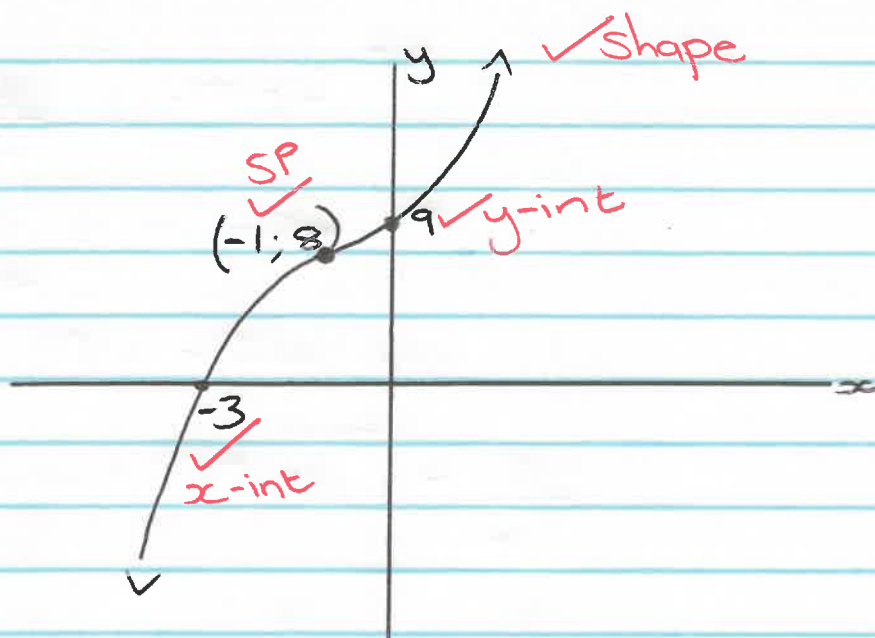
$$\therefore m = -4$$

$$y + 2 = -4(x - 1)$$

$$y = -4x + 2 \checkmark$$

(4)

1.4.



(4)

y-int:

$$y = (0+1)^3 + 8$$
$$= 9$$

x-int

$$0 = (x+1)^3 + 8$$

$$(x+1)^3 = -8$$

$$x+1 = -2$$

$$x = -3$$

Question 2

2.1. $D(0; -4)$ ✓

(1)

2. $f'(x) = 3ax^2 + b$ ✓

$$f'(1) = 3a + b$$

$$\therefore 3a + b = 0 \dots \textcircled{1} \text{ ✓ } f'(1) = 0$$

$$f(1) = a + b - 4$$

$$a + b - 4 = 0 \dots \textcircled{2} \text{ ✓ } f(1) = 0$$

$$\textcircled{1} - \textcircled{2}$$

$$2a + 4 = 0$$

$$2a = -4$$

$$a = -2$$

✓^m Can also use substitution

$$\therefore -2 + b - 4 = 0$$

$$b = 6$$

$$\therefore f(x) = -2x^3 + 6x - 4$$

(4)

$$2.3. f(x) = -2x^3 + 6x - 4$$

$$f'(x) = -6x^2 + 6 \quad \checkmark$$

$$-6x^2 + 6 = 0 \quad \checkmark f' = 0$$

$$-6x^2 = -6$$

$$x^2 = 1$$

$$x = 1$$

N/A

or

$$x = -1 \quad \checkmark$$

$$\therefore y = -8$$

$$\therefore A(-1; -8) \quad \checkmark$$

Must be coordinate

(4)

$$2.4. -2x^3 + 6x - 4 = 0 \quad \checkmark f = 0$$

$$x^3 - 3x + 2 = 0$$

$$(x-1)(x^2+x-2) = 0$$

$$(x-1)(x-1)(x+2) = 0$$

$$x = 1 \quad \text{or} \quad x = -2 \quad \checkmark$$

$$\therefore C(-2; 0) \quad \checkmark$$

(3)

$$2.5. f''(x) = -12x \quad \checkmark$$

$$f''(0) = -12(0)$$

$$= 0 \quad \checkmark f'' = 0$$

$$f''(0) = 0$$

(2)

6.1. $x \leq -1$ ✓ or $x \geq 1$ ✓ (2)

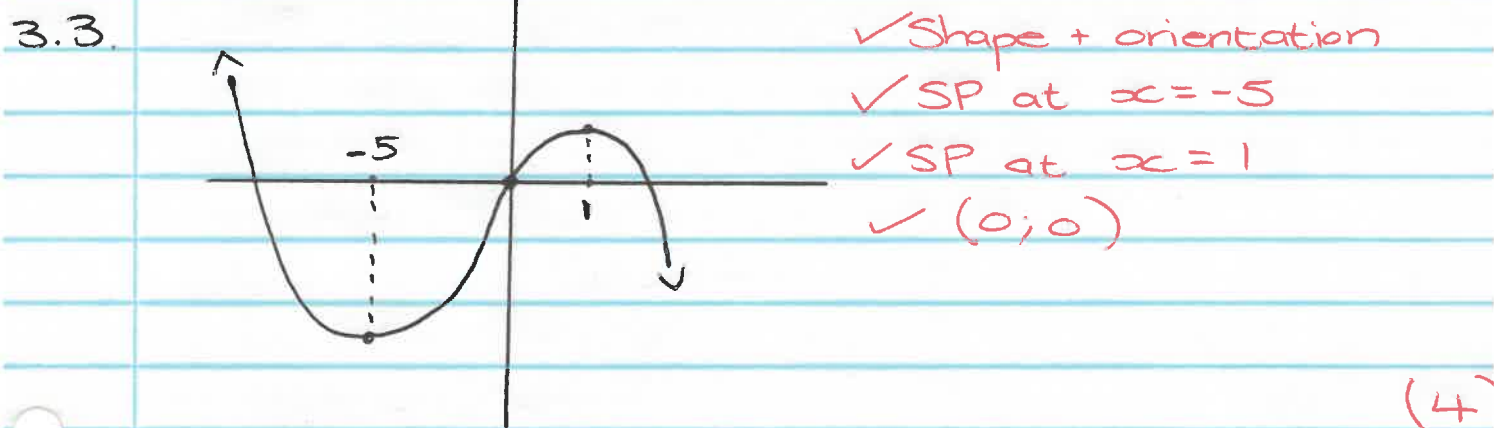
6.2. $x < -1$ ✓ or $0 < x < 1$ ✓ (2)

2.7. $k = -8$ ✓^{CA} or $k = 0$ ✓ (2)

Question 3

3.1. 15 ✓ (1)

3.2. $x = -2$ ✓✓ (2)



3.4. $x < -2$ ✓^{CA} (2)

Question 4

$$\begin{aligned} 4.1. \text{ Area} &= AB \times BC \checkmark \\ &= x \times (-3x + 12) \checkmark \\ &= -3x^2 + 12x \end{aligned} \quad (2)$$

$$4.2. A' = -6x + 12 \checkmark$$

$$\begin{aligned} -6x + 12 &= 0 \quad \checkmark A' = 0 \\ -6x &= -12 \\ x &= 2 \checkmark \end{aligned}$$

$$\begin{aligned} \therefore A(2) &= -3(2)^2 + 12(2) \\ &= 12 \checkmark \end{aligned}$$

$$\therefore \text{Max area is } 12 \text{ units}^2 \quad (4)$$

Later